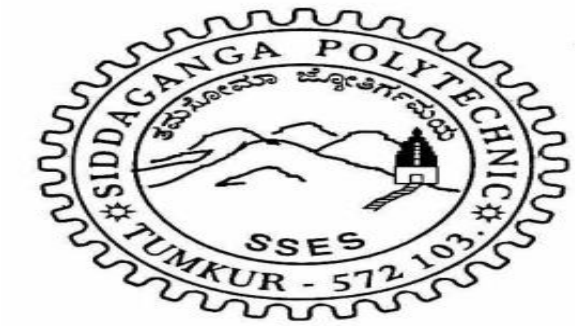
Siddaganga Polytechnic

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Department of Computer Science & Engineering



FULL STACK DEVELOPMENT (20CS52I)

A Lab Manual Submitted By

5th Semester

DEPT. OF COMPUTER SCIENCE AND ENGINEERING

UNDER THE GUIDANCE OF

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Siddaganga Polytechnic.

Tumkur-3

FULL STACK DEVELOPMENT

List of Experiments:-

Sl.no	Experiments
1	Create User Stories ,Product backlog, workflow and start sprint using appropriate tool like Jira
2.	Creation of local Repository on Git and push it into Remote Repository and Working with Git Commands
3.	Setting up React environment and Create and running a React.js app
4.	Setting Up the Environment and Tools for front-end development Installing VS Code
5.	Create a form like registration form or feedback form, after submit hide create form and enable the display section using java script.
6.	Create and run simple program in Typescript
7.	Create and run a simple Maven project
8.	Create a Spring Boot Project in Eclipse IDE
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11.	Download and Install MongoDB and perform CRUD operation on database
12.	Create and Run Servlet class using Tomcat server
13.	Download and Install Docker & Execute simple commands in Linux

1. Create User Stories ,Product backlog, workflow and start sprint using appropriate tool like Jira

How to create project plan and product backlog for project and User story creation.

- Open browser, search for Jira Login.
- Continue with your Gmail account or login to Jira.
- Click on Jira software and select project from top menu bar then select create project from dropdown Menu.
- Select Scrum click on template and click on create.
- Give a name to your project and Give a Description if you want .
- Click on create.
- Select issues from top menu bar and select issue type .This will be default setting.
- Give a summary to your project.
- Now write a user story in Description box.
- Your story will then go into the backlog to be Assigned and auctioned by the project manager, product owner or other relevant stakeholders and click on start sprint
- Click on Board and select Insights
- Click on Insights and click “Sprint burn down”
- And click on Learn more.

1. Create and manage product backlog using appropriate tool like Jira

Summary: Customer registration functionality

Description

AS A customer

I WANT to have registration functionality

SO THAT I can successfully resist

Scope

- build a registration page
- customer validation

- customer should be able to change the phone number
- it should work in all the browser
- it should also work in mobile

Pre condition

- customer should have email and phone number

Acceptance criteria

Scenario 1: customer can successfully resister

- "Given" I am on registration page
- "And" I give valid customer name and phone number
- "And" I check on sign in
- "Then" I will successfully resister

Scenario 2: customer cannot successfully resister

- "Given" I am on registration page
- "And" I give invalid customer name and phone number
- "Then" I will get a error message as "registration failed incorrect customer name"

Summary: Customer checking availability

Description

AS A customer

I WANT to have checking available of hall

SO THAT i can check the available halls

Scope

- build a available checking page
- it should be only inside the Karnataka
- customer should be able to check the available halls in their particular location

pre condition

- customer should have nearest halls in their location

Acceptance criteria

Scenario 1: Customers can successful check availability of hall in their location

- "Given " I am on check available of hall page
- "And" I give particular location and date

Scenario 2: customer can't successfully check availability of hall in their location

- "Given " I am on check available of hall page
- "And" I give wrong location
- "Then" I will get the error message as in valid location

Summary: Customer booking hall

Description

AS A customer

I WANT to booking hall

SO THAT i can book the hall

Scope

- build a booking hall page
- customer should be able to change the date and location

Pre condition

- customer should be able to book the hall in their particular date

Acceptance criteria

Scenario 1: customer can successfully booking hall

- "Given" I am on booking page
- "And" I give available date time
- "And" I will book the hall
- "Then" I successfully booked the hall

Scenario 2: customer can't successfully booking hall

- "Given" I am on booking page
- "And" I give invalid date and time
- "Then "I will get the error messages as their hall is already booked

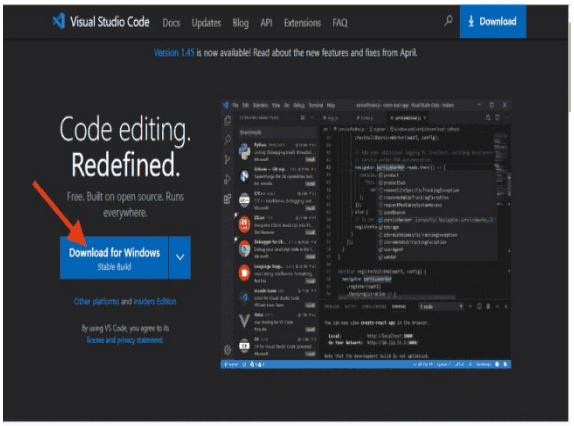
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Summary: Customer booking details

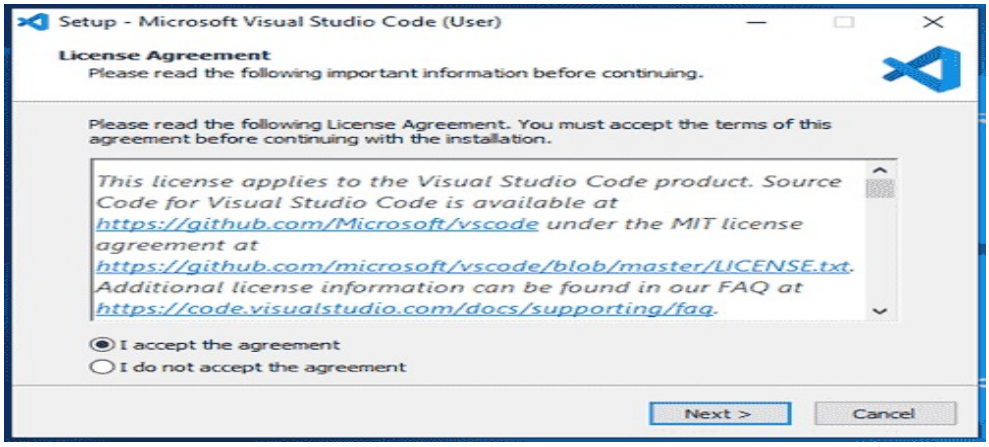
Description:

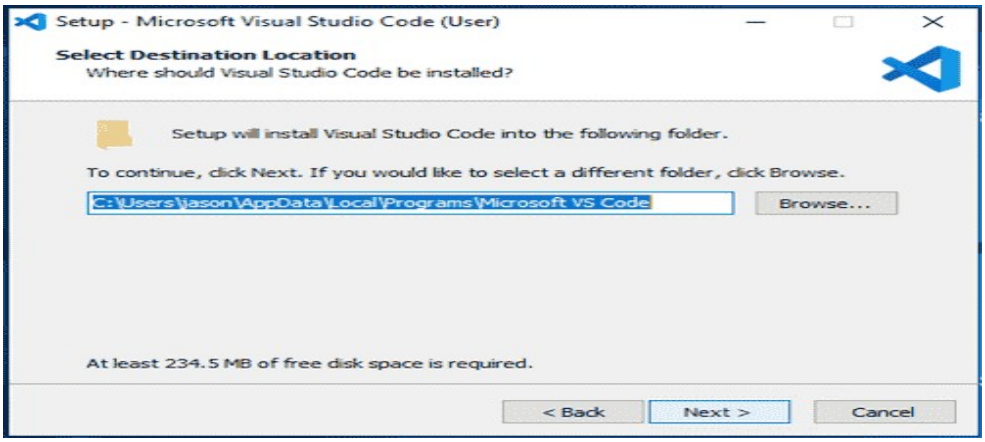
AS A customer

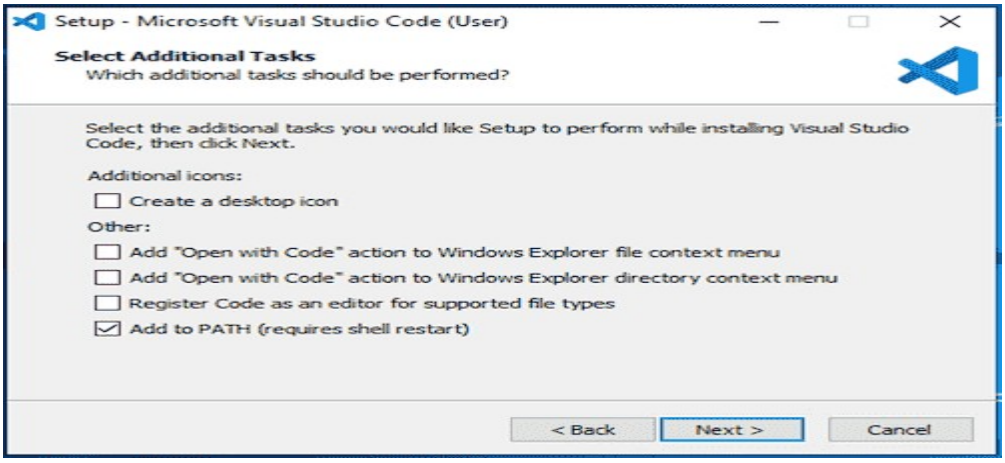
I WANT to block the hall

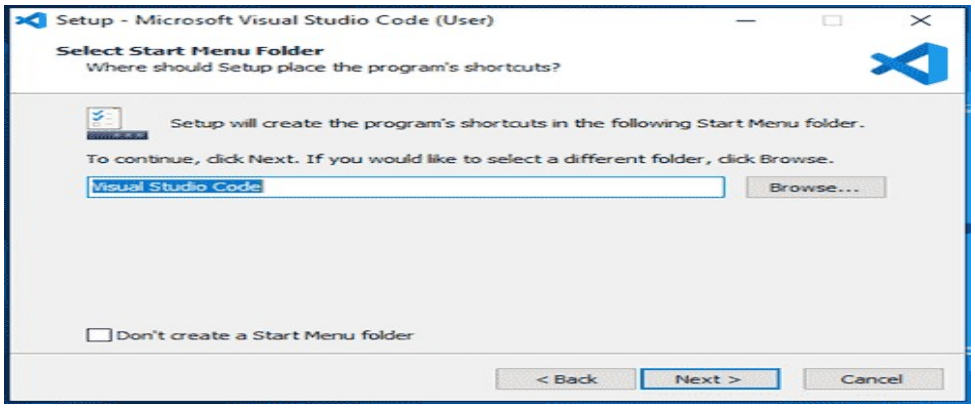


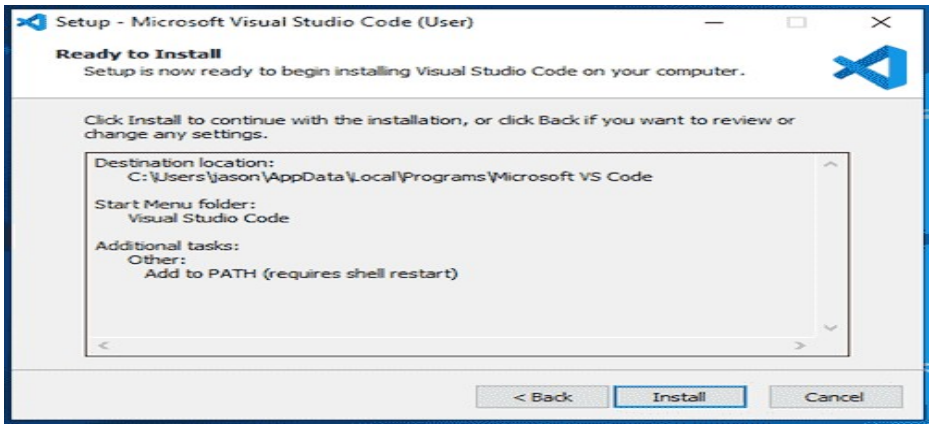
Install Visual Studio Code by opening the downloaded setup file and following the prompts.

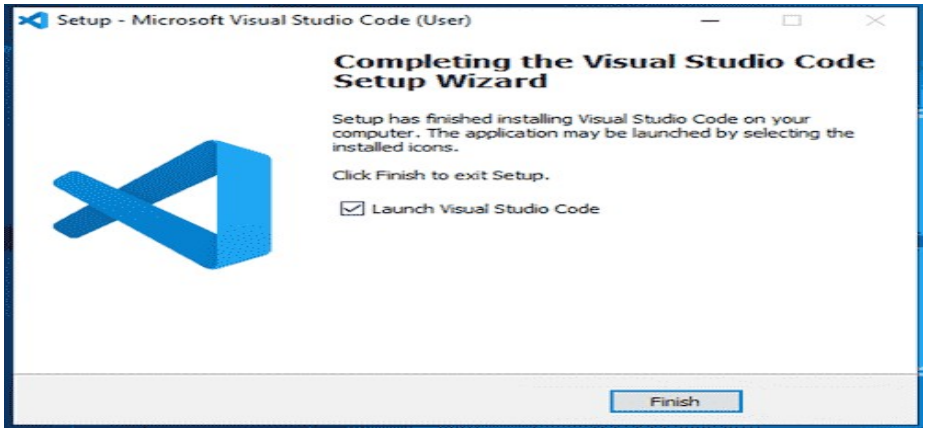



Setup - Microsoft Visual Studio Code (User)
Select Destination Location
Where should Visual Studio Code be installed?
Setup will install Visual Studio Code into the following folder.
To continue, click Next. If you would like to select a different folder, click Browse.
C:\Users\jason\AppData\Local\Programs\Microsoft VS Code
Browse...
At least 234.5 MB of free disk space is required.
< Back
Next >
Cancel


Setup - Microsoft Visual Studio Code (User)
Select Additional Tasks
Which additional tasks should be performed?
Select the additional tasks you would like Setup to perform while installing Visual Studio Code, then click Next.
Additional icons:
Create a desktop icon
Other:
Add "Open with Code" action to Windows Explorer file context menu
Add "Open with Code" action to Windows Explorer directory context menu
Register Code as an editor for supported file types
Add to PATH (requires shell restart)
< Back
Next >
Cancel


Setup - Microsoft Visual Studio Code (User)
Select Start Menu Folder
Where should Setup place the program's shortcuts?
Setup will create the program's shortcuts in the following Start Menu folder.
To continue, click Next. If you would like to select a different folder, click Browse.
Visual Studio Code
Browse...
Don't create a Start Menu folder
< Back
Next >
Cancel





5.Create a form like registration form or feedback form, after submit hide create form and enable the display section using java script.

Registration.html

```
<html> <head>
    <title> Registration Form</title>
    <script>
      function passvalues()
      {
            var name = document.getElementById("name").value;
            var email = document.getElementById("email").value;
            var address = document.getElementById("address").value;
            localStorage.setItem("name",name);
            localStorage.setItem("email",email);
            localStorage.setItem("address",address);
            return;
      }
```

```
        </script>
    </head>
    <body>
<h1>Registrtion Form</h1>
        <form action="Details.html">
<fieldset>
    <legend>Registration</legend>
 <label> Name </label>
        <input type="text" id="name"/><br><br>
 <label> Email ID </label>
        <input type="email" id="email"/><br><br>

 <label> Address </label>
        <input type="address" id="address"/><br><br>
        <input type="submit" value="submit" onclick="passvalues()"/>
</fieldset>
        </form>
    </body>
</html>
```

Details.html

```
<html>
    <head>
        <title> Details</title>
    </head>
    <body>
<form>
    Your Name is:<p id="name"></p><br>
        Your email is:<p id="email"></p><br>
        Your address is:<p id="address"></p>
<script>
            document.getElementById("name").innerHTML = localStorage.getItem("name");
            document.getElementById("email").innerHTML = localStorage.getItem("email");
            document.getElementById("address").innerHTML = localStorage.getItem("address");

        </script>
</form>

    </body>
</html
```

6. Create and run simple program in TypeScript

Install TypeScript using Node.js Package Manager (npm)

Step-1 Install Node.js. It is used to setup TypeScript on our local computer.

To install Node.js on Windows, go to the following link: **https://www.javatpoint.com/install-nodejs**

Step-2 Install TypeScript. To install TypeScript, enter the following command in the Terminal Window.

- npm install typescript --save-dev //As dev dependency
- npm install typescript -g //Install as a global module

  or
- npm install -g typescript
- npm install typescript@latest -g //Install latest if you have an older version

Step-3 To verify the installation was successful, enter the command **$ tsc -v** in the Terminal Window.

Install Live server

```
npm install -g live-server
```

Create and run first program in TypeScript

- open command prompt
- go to d: drive(any drive)
- d:\>mkdir typescript
- d:\>cd typescript
- d:\typescript> npm install typescript --save-dev
- open visual studio code
- file-open folder-choose typescript folder from d:
- create new file- save it as types.ts(any name.ts)
- Write the below code and save it
- console.log("Hello World");
- go to command prompt and compile the program

- tsc types.ts
- run the program
- node types.js
- Observe the output

Example:

if-else:

```
var age=18;
if(age>=18)
{
 console.log("You are eligible for voting");
}
else{
 console.log("You are not eligible for voting");
}
```

OR

while:

```
let counter = 0;
while (counter < 5) {
 console.log(counter);
 counter++;
}
```

7. Create and run simple program in Apache Maven

- Maven is a Java based tool, so the very first requirement is to have JDK installed on your machine.

➢ To install maven on windows, you need to perform following steps:

- ✔ Download maven zip file and extract it.
- ✔ Add JAVA_HOME in user variables environment.
- ✔ Add maven bin directory and JDK path to system variables environment.

✔ Verify Maven in **CMD** by the following command.

- mvn –version

- Open Eclipse. Select File > Other > Maven. Select Maven Projects Option. Click on Next Button.
- Select create a simple project and click next
- Give group id and artifact id and click on finish
- Then run a simple java program.

```
public class Demo{
        public static void main(String[] args) {
        int a = 3;
        int b = 5;
        int sum = a+b
        System.out.println("The sum is"+sum);
        }
}
```

8. Create a Spring Boot Project in Eclipse IDE

- Open the Spring Initializr https://start.spring.io/.
- Select the Spring Boot version 3.0.11 BUILD-SNAPSHOT.
- Provide the Group name. We have provided the Group name com.javatpoint.
- Provide the Artifact. We have provided the Artifact spring-bootapplication-run.
- Add the Spring Web dependency.
- Click on the Generate button. When we click on the Generate button, it wraps all the specifications related to application into a Jar file and downloads it to the local system.
- Extract the jar file.
- Go to file > Open Projects from file system > Browse extracted directory import it into import source.
- And at last click on the Finish button and we are done creating the Spring Boot project.
- By now, Spring Boot project has been created as depicted in the below media.

Spring Version:

Step 1: Install Eclipse IDE for Enterprise Java and Web Developer Please refer to this article How to Install Eclipse IDE for Enterprise Java and Web Development and install the Eclipse IDE.

Step 2: Create a Spring Boot Project in Spring Initializr Go to this link and create a Spring Boot project. Please fill in all the details accordingly and at last click on the GENERATE button below. This will download your Spring Boot project in zip format. Now extract the folder into your local machine.

Step 3: Import Spring Boot Project in Eclipse IDE Go to the Eclipse IDE for Enterprise Java and Web Developer > File > Import as shown in the below image.

Step 4: Search "maven" and choose Existing Maven Project and click on the Next button as shown in the below image.

Step 5: Now click on the Browse button and select the extracted zip file that has been generated. Step 6. And at last click on the Finish button and we are done creating the Spring Boot project.

By now, Spring Boot project has been created as depicted in the below media

9. Initialize Spring boot Project

```
package com.example.demo;
import java.util.*;

import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;

@SpringBootApplication
public class DemoApplication {
	public static void main(String[] args) {
		Scanner c = new Scanner(System.in);
		System.out.println("enter the number for a=");
		int a= c.nextInt();
```

```
                System.out.println("enter the number for b");
                int b= c.nextInt();
                System.out.println(a+b+" is the sum");
        }
}
```

10. Download And Install MySql and perform CRUD operation in MySql.

- Open the MySQL website on any Browser.
- Select the Download option.
- Select MySQL Installer for Windows
- Choose the desired installer and click on download.
- After download open installer.
- Click Yes, it asks to choose setup area. Select custom and Click Next.
- Open MySQL. Select the server you want to install, and move it to products button.
- Click on Next > execute to download and install MySQL Server, shell and workbench
- Once the product is ready to configure, click Next. Under Type and Networking. Click Next.
- For authentication use password
- Select MySQL on Most Root password and click on next.
- Go for default windows, under apply configuration click on execute, and click on finish.
- Complete the installation. This will launch MySQL Workbench and shell.

CRUD Operation in MySQL

1) Create operation:

```
CREATE DATABASE college;
USE college;

CREATE TABLE student(id int primary key ,Name varchar(25),Age int,Address varchar(27));

INSERT INTO student VALUES(1,"Raghav",18,"Tumkur"),
(2,"punith",19,"chikkamanglore"),(3,"udaykumar",15,"benglore");
```

2) Read operation

```
SELECT * FROM student;
SELECT Name,Address FROM student WHERE id<=2;
```

3) Update operation:

```
UPDATE student set Name="Bhavyamam" WHERE id=3;
UPDATE student set Age=38 WHERE id=3;
```

4) Delete operation:

```
DELETE FROM student WHERE id=1;
DELETE FROM student WHERE id=3;
```

11. Download and Install MongoDB and perform CRUD operation on database

1. Go to browser and download mongoDb community server (https://www.mongodb.com/try/download/community a)
2. Once download is complete open the msi file. Click Next in the start up screen
3. Accept the End-User License Agreement > Click Next
4. Click on the "complete" button
5. Click on the "complete" button to install all of the components.
6. Select "Run service as Network Service user". make a note of the data directory, we'll need this later > Click Next
7. Click on the Install button to start the installation.
8. Once complete the installation, Click on the Finish button.

MangoDB Shell

1. To install the new MongoDB shell (mongosh), visit www.mongodb.com and click on Product menu -> Tools -> Database Shell, as shown below.
2. On the MongoDB Shell page, click on the Download button to download the shell.
3. Now, click on the downloaded installer file to start the installation wizard and click on next.
4. Click on the Install button to start the installation. It should quickly install it click on finish button
5. Click on the mongosh.exe to open a new MongoDB shell and Press Enter to start the shell,

CRUD Operation in MongoDB Shell

- Create or insert Data

```
> show dbs
admin   0.000GB
config  0.000GB
local   0.000GB
zain    0.000GB
>
```

```
> db.data.insertOne({name:"zain", age:21, degree:"BSCS"})
{
        "acknowledged" : true,
        "insertedId" : ObjectId("5fde0df4d88d72bf3d337c79")
}
>
```

- Read in mongoDB

```
> db.data.find().pretty()
{
        "_id" : ObjectId("5fde0df4d88d72bf3d337c79"),
        "name" : "zain",
        "age" : 21,
        "degree" : "BSCS"
}
{
        "_id" : ObjectId("5fde0eaed88d72bf3d337c7a"),
        "name" : "usman",
        "age" : 25,
        "degree" : "BSSE"
}
{
        "_id" : ObjectId("5fde0eaed88d72bf3d337c7b"),
        "name" : "joe",
        "age" : 35,
        "degree" : "BA"
}
>
```

❖ **Update in mongoDB**

```
> db.data.updateOne({name:"zain"},{$set :{age:22}})
{ "acknowledged" : true, "matchedCount" : 1, "modifiedCount" : 1 }
> db.data.find().pretty()
{
        "_id" : ObjectId("5fde0df4d88d72bf3d337c79"),
        "name" : "zain",
        "age" : 22,
        "degree" : "BSCS"
}
{
        "_id" : ObjectId("5fde0eaed88d72bf3d337c7a"),
        "name" : "usman",
        "age" : 25,
        "degree" : "BSSE"
}
{
        "_id" : ObjectId("5fde0eaed88d72bf3d337c7b"),
        "name" : "joe",
        "age" : 35,
        "degree" : "BA"
}
```

❖ **Deletion with mongoDB**

```
> db.data.find().pretty()
{
        "_id" : ObjectId("5fde0df4d88d72bf3d337c79"),
        "name" : "zain",
        "age" : 22,
        "degree" : "BSCS"
}
{
        "_id" : ObjectId("5fde0eaed88d72bf3d337c7a"),
        "name" : "usman",
        "age" : 25,
        "degree" : "BSSE"
}
{
        "_id" : ObjectId("5fde0eaed88d72bf3d337c7b"),
        "name" : "joe",
        "age" : 35,
        "degree" : "BA"
}
{
        "_id" : ObjectId("5fde1bbdd88d72bf3d337c7c"),
        "name" : "zunair",
        "age" : 21,
        "degree" : "BSCS"
}
```

```
> db.data.deleteOne({degree:"BA"})
{ "acknowledged" : true, "deletedCount" : 1 }
> db.data.find().pretty()
{
        "_id" : ObjectId("5fde0df4d88d72bf3d337c79"),
        "name" : "zain",
        "age" : 22,
        "degree" : "BSEE"
}
{
        "_id" : ObjectId("5fde0eaed88d72bf3d337c7a"),
        "name" : "usman",
        "age" : 25,
        "degree" : "BSSE"
}
{
        "_id" : ObjectId("5fde1bbdd88d72bf3d337c7c"),
        "name" : "zunair",
        "age" : 21,
        "degree" : "BSEE"
}
```

12. Create and Run Servlet class using Tomcat server

- **Apace Tomcat**

- Open any Browser and search Apache Tomcat website, click on Tomcat 10 and download 32-bit/64-bit Windows Service Installer
- Go to downloads, open the file and click next to proceed > read the license agreement and click on I agree > in components selection select full and click next.
- Give the username and password and click next > and select where the folder should be installed and click on install.
- Open Eclipse, go to help > install new software > in (work – with) box select all available sites.
- Click on Web, XML, Java EE and OSGi Enterprise development and click next > accept the agreement > and click on finish.
- It will ask to restart eclipse, click on restart.
- Click on open perspective on the top right corner of eclipse > select Java EE and click open.
- Go to server and click on add server > in define a new server click on apace > select latest version of tomcat server and click next, here browse for the directory you need to install the server > OK > Finish.
- Go to file > new > Dynamic web project > Enter the project name and keep the rest of the settings as it is and click next, select Generate web.xml deployment descriptor and click on finish, web project will be generated.
- Create a servlet class, go to file > new > servlet. Enter java package name and class name, click next > Deployment descriptor (web.xml) can be specified if required > click next and finish.
- Configurations will be generated in eclipse editor, change the description as required and run the servlet code.

- To run the code right click on the servlet > run as > run on server > select to run on existing server and select the server on which the code should be executed > next > finish.
- To restart the server and to run the application click ok and output will be displayed.

13. Download and Install Docker & Execute simple commands in Linux.

Docker Installation Steps

1. For installation of Docker, go to browser and search for Docker download. And click on the first link that appears.
2. After clicking the link, choose your OS, be it Windows, MAC, or Linux for installation.
3. After installation, Open the Docker Desktop.
4. Accept the Docker Subscription Service Agreement window. and click On continue. Docker Desktop will start after we accept the terms and conditions.
5. After installation it will ask to restart your computer click on restart.
6. Open docker desktop then run git bash as administrator and execute the following commands.

Commands for docker

- To verify docker is installed or not
  - docker
- To check version of docker
  - docker version
- To run built in container of docker
  - docker run hello-world
- To run ubuntu in docker
  - docker pull ubuntu
  - docker run -it -d ubuntu
  - docker ps -a
  - winpty docker exec -it <id of ubuntu container> bash

    # mkdir spt // to create directory

    # cd spt // to change directory

```
# cat > demo                    // to create a file
# cat demo (or) echo demo       // to display contents of file
# rm demo                       // to delete a file
# rmdir spt                     // to delete directory
```

- To stop the running container
  - docker stop <Container Id>
- To remove the running container
  - docker remove <Container Id>